

BLACKBOX X — PHASE 3.8

MONTE CARLO & PROBABILISTIC RISK INTELLIGENCE
ARCHITECTURE — Complete Mathematical, Algorithmic, and
Technical Specification — 1.1 Objective & Research Framing — 1.2
Institutional Positioning & Non-Predictive Charter

DOCUMENT
MONTE-CARLO-ARCHITECTURE.pdf

SOURCE
**docs/MONTE-CARLO-
ARCHITECTURE.md**

SYSTEM
BLACKBOX X Engine

STATUS
Documentation

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BLACKBOX X — PHASE 3.8

MONTE CARLO & PROBABILISTIC RISK INTELLIGENCE ARCHITECTURE

Complete Mathematical, Algorithmic, and Technical Specification

1. Purpose & Philosophy

1.1 Objective & Research Framing

Phase 3.8 extends BLACKBOX X from deterministic historical backtesting and discrete simplex portfolio optimization into **probabilistic simulation and distribution intelligence**.

The core quantitative research question answered by this engine is:

"Given the observed return characteristics and a defined simulation methodology, what distribution of outcomes appears across reproducible simulated paths?"

1.2 Institutional Positioning & Non-Predictive Charter

BLACKBOX X explicitly **does not predict future markets**.

Monte Carlo simulations do not generate market forecasts, expected payoffs, or price targets. Instead, the engine evaluates the mathematical dispersion, path-dependency, recovery dynamics, and extreme downside tail risks of a portfolio under explicitly stated, transparent probabilistic assumptions.

Core Mandate:

- **No Forecast Claims:** Outputs represent simulated distributions under declared assumptions, not future probabilities of real-world returns.
- **Strict Methodological Transparency:** Every simulation run must carry full provenance regarding its sampling mechanics, seed, distributional assumptions, and underlying historical baseline.
- **Reproducibility:** The engine uses a deterministic, seeded pseudo-random number generator (PRNG). Identical inputs and seeds produce bit-identical simulation paths and summary metrics.

2. Existing Architecture Reuse & Non-Negotiable Boundaries

Phase 3.8 builds directly upon the validated foundations of Phases 1 through 3.7 without modifying or duplicating existing engines.

BLACKBOX X PLATFORM	
Phase 1 & 2:	Deterministic Data Engine (1,826 daily bars, 1,825 returns)
Phase 3.1:	Research Trail & Ghost Mode Insight Engine
Phase 3.2:	Multi-Scenario Stress Lab (Macro & Historical Shocks)
Phase 3.3:	Strategy Genome (Graph Topology & Phenotype Metrics)
Phase 3.4:	Institutional Risk Committee (ResearchPack & Briefings)
Phase 3.5:	AI Research Assistant (Featherless Inference & Tool Sandbox)
Phase 3.6:	Autonomous Quant Research Agent (Hypothesis & Contradiction)
Phase 3.7:	Portfolio Covariance, Simplex Optimizer, Euler Risk, Backtest
PHASE 3.8:	MONTE CARLO & PROBABILISTIC RISK INTELLIGENCE (NEW ENGINE) - Joint Historical Bootstrap (Vector Resampling) - Parametric Correlated Normal (Cholesky Decomposition) - Path Drawdown & Terminal Wealth Distributions - Web Worker High-Throughput Streaming Engine

2.1 Reused Components (Zero Code Duplication)

- 1. **Asset Universe & Data Engine (`src/core/data.ts`):**
 - The tri-asset universe (`GOLD` , `BTC` , `NVDA`).
 - The 1,826 daily closing price bars spanning `2019-01-01` to `2023-12-31` .
- 2. **Synchronized Covariance & Return Matrix (`src/core/portfolio/covariance.ts`):**
 - The 1,825 synchronized daily observations vector $T = 1,825$.
 - Empirical mean return vector μ and daily covariance matrix Σ_{daily} .
 - Portfolio weight validation and simplex projection ($\mathbf{w} \in \Delta^2$).
- 3. **Transaction Cost & Rebalancing Logic (`src/core/portfolio/portfolioBacktest.ts`):**
 - Next-bar rebalancing schedules (`BUY_AND_HOLD` , `DAILY` , `MONTHLY` , `QUARTERLY` , `THRESHOLD \pm 5\%`).
 - Standard 10 bps friction and capital turnover accounting.
- 4. **Risk Metrics & VaR Framework (`src/core/portfolio/varMetrics.ts`):**
 - Positive loss convention $L = -R$.
 - Non-parametric historical quantile calculations.
- 5. **Research Governance & Integrity (`src/core/researchAgent/`):**
 - Evidence collection contracts (`evidenceId` , direct evidence vs. interpretation).
 - Strict tool execution budget (bounded research steps).

3. Data Source, Calendar Alignment & Provenance

3.1 Source Dataset Characteristics

- **Data Source:** Offline deterministic simulated demonstration dataset.
- **Date Range:** `2019-01-01` to `2023-12-31` .
- **Price Observations:** Exactly `1,826` daily closing price observations per asset.
- **Synchronized Observations:** Exactly `1,825` contemporaneous daily return intervals.
- **Terminology:** Must strictly use "**synchronized daily observations**". It is forbidden to characterize these as "actual market trading sessions" or "real historical exchange feeds".

3.2 Mandatory Provenance Metadata

Every Monte Carlo simulation result, export, and AI tool payload must serialize the following provenance record:

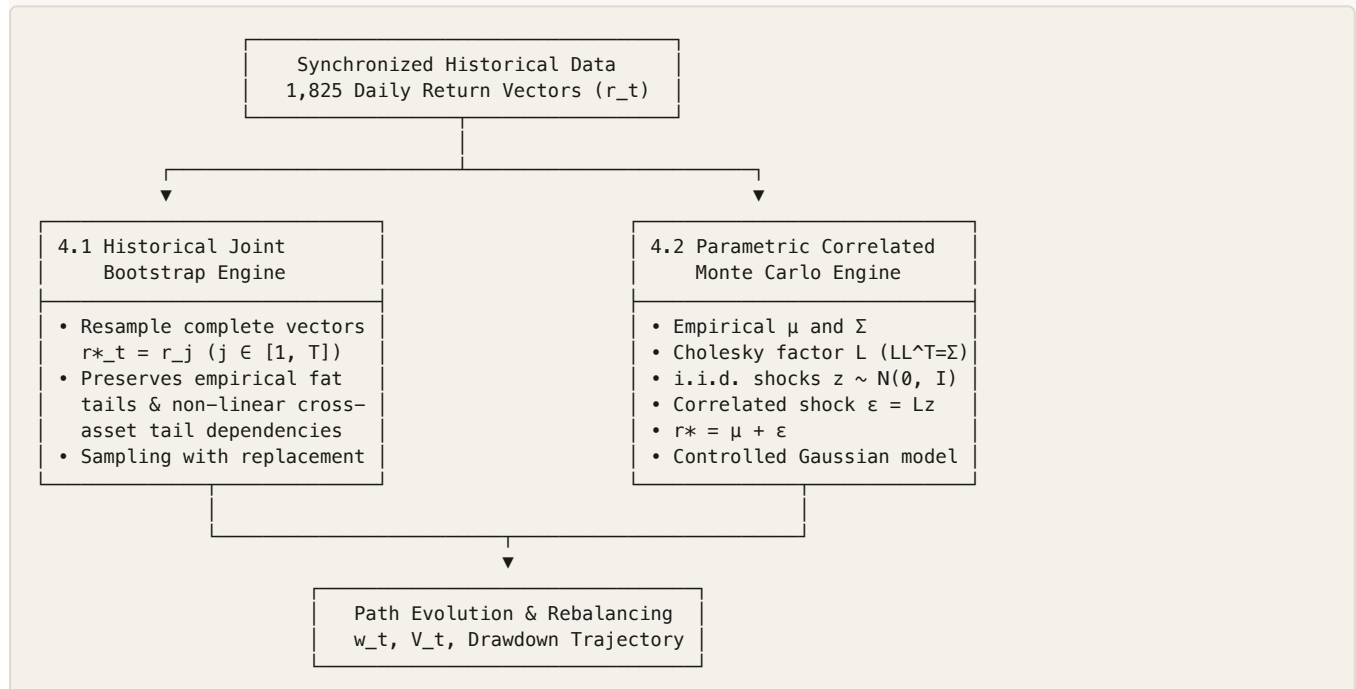
```

export interface MonteCarloProvenance {
  dataSource: 'OFFLINE_DEMO';
  dataSourceLabel: string;           // 'Offline Simulated Demonstration Dataset'
  dataWindow: {
    startDate: '2019-01-01';
    endDate: '2023-12-31';
    observationCount: 1825;
  };
  simulationMethod: MonteCarloMethod; // 'HISTORICAL_BOOTSTRAP' | 'PARAMETRIC_NORMAL'
  simulationCount: number;             // e.g., 10,000
  horizonDays: number;                // e.g., 252
  seed: number;                       // Deterministic PRNG seed
  portfolioWeights: PortfolioWeights; // { GOLD: number, BTC: number, NVDA: number }
  rebalanceSchedule: RebalanceFrequency;
  transactionCostBps: number;
  fingerprint: string;               // Deterministic SHA-256 hash
  generatedTimestamp: number;
}

```

4. Simulation Methodologies & Mathematics

Phase 3.8 specifies **two primary simulation engines**, each addressing distinct risk-modeling assumptions.



4.1 Method A: Historical Joint Bootstrap

The joint bootstrap generates simulated paths by resampling entire multi-asset return vectors directly from the synchronized historical observation matrix.

Mathematical Formulation

Let the synchronized historical dataset be represented as a matrix $\mathbf{R} \in \mathbb{R}^{T \times N}$, where $T = 1,825$ and $N = 3$:

$$\mathbf{r}_t = \begin{bmatrix} r_{\text{GOLD},t} \\ r_{\text{BTC},t} \\ r_{\text{NVDA},t} \end{bmatrix}, \quad t \in \{1, \dots, T\}$$

For each simulated path $m \in \{1, \dots, M\}$ and each simulated forward day $\tau \in \{1, \dots, H\}$:

1. Draw a discrete uniform random index j from the historical index set:

$$j \sim \mathcal{U}\{1, T\}$$

2. Select the complete synchronized return vector at index j :

$$\mathbf{r}_{m,\tau}^* = \mathbf{r}_j = \begin{bmatrix} r_{\text{GOLD},j} \\ r_{\text{BTC},j} \\ r_{\text{NVDA},j} \end{bmatrix}$$

Critical Invariant: Complete Vector Resampling

- **Zero Asset-Decoupling:** It is **strictly prohibited** to bootstrap assets independently ($r_{\text{GOLD},\tau}^* = r_{\text{GOLD},j_1}$, $\{\text{BTC}, \tau\} = r_{\{\text{BTC}, j_2\}}$, $r_{\text{NVDA},\tau}^* = r_{\text{NVDA},j_3}$ with $j_1 \neq j_2 \neq j_3$).
- **Rationale:** Independent resampling annihilates the empirical cross-asset covariance, joint crash risk, and flight-to-safety dynamics (e.g., Gold appreciation during crypto/equity drawdowns).
- **Sampling Convention:** Sampling is performed **with replacement**.

4.2 Method B: Parametric Correlated Monte Carlo

The parametric simulation generates multivariate normal innovations that preserve the empirical first and second moments of the asset universe.

Mathematical Formulation

1. Empirical Moments:

- Daily mean return vector: $\boldsymbol{\mu} \in \mathbb{R}^N$

$$\mu_i = \frac{1}{T} \sum_{t=1}^T r_{i,t}$$

- Daily sample covariance matrix: $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$

$$\Sigma_{ij} = \frac{1}{T-1} \sum_{t=1}^T (r_{i,t} - \mu_i)(r_{j,t} - \mu_j)$$

2. **Cholesky Factorization:** Compute the unique lower-triangular matrix $\mathbf{L} \in \mathbb{R}^{N \times N}$ such that:

$$\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$$

where:

$$L_{j,j} = \sqrt{\Sigma_{j,j} - \sum_{k=1}^{j-1} L_{j,k}^2}$$

$$L_{i,j} = \frac{1}{L_{j,j}} \left(\Sigma_{i,j} - \sum_{k=1}^{j-1} L_{i,k} L_{j,k} \right), \quad i > j$$

3. **Standard Normal Innovations:** Draw an independent and identically distributed standard normal vector $\mathbf{z} \in \mathbb{R}^N$ using the Box-Muller transform or Marsaglia polar method over deterministic PRNG uniform draws:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

4. **Correlated Shock Transformation:** Transform $\mathbf{z}$ into correlated shocks $\boldsymbol{\varepsilon} \in \mathbb{R}^N$:

$$\boldsymbol{\varepsilon} = \mathbf{L}\mathbf{z}$$

Since $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T] = \mathbf{L}\mathbb{E}[\mathbf{z}\mathbf{z}^T]\mathbf{L}^T = \mathbf{L}\mathbf{I}\mathbf{L}^T = \boldsymbol{\Sigma}$, the simulated vector preserves cross-asset covariance.

5. **Simulated Return Vector:**

$$\mathbf{r}_{m,\tau}^* = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}_{m,\tau}$$

Numerical Stabilization Policy for Non-Positive-Definite Covariance

Due to finite floating-point precision, empirical covariance matrices can occasionally fail strict positive-definiteness ($L_{j,j}^2 \leq 0$).

Stabilization Algorithm (Reactive, Not Preemptive):

1. Attempt standard Cholesky decomposition on the unadjusted empirical covariance matrix $\boldsymbol{\Sigma}$.
2. **Conditional Activation:** Regularization must **only activate if and when** the original decomposition encounters a non-positive pivot ($L_{j,j}^2 \leq 0$ or $L_{j,j}^2 < \epsilon$, where $\epsilon = 10^{-10}$). Regularization is **never applied preemptively**.
3. If decomposition fails:
 - Apply deterministic Tikhonov regularization (minimal diagonal jitter):

$$\boldsymbol{\Sigma}_{\text{stable}} = \boldsymbol{\Sigma} + \lambda \mathbf{I}$$

where $\lambda = \max(10^{-8}, -\min(\text{eig}(\boldsymbol{\Sigma})) + 10^{-8})$.

- Re-attempt Cholesky decomposition on $\boldsymbol{\Sigma}_{\text{stable}}$.
4. Log the regularization event in `MonteCarloResult.stabilizationApplied = true` with the exact adjustment magnitude λ . If standard decomposition succeeds without regularization, `stabilizationApplied = false` and λ is omitted.
 5. **Prohibition:** Silent manipulation of off-diagonal covariances is forbidden.

5. Random Number Generation & Deterministic PRNG

5.1 Rejection of `Math.random()`

`Math.random()` is **strictly banned** from the simulation engine. It is non-deterministic, browser-dependent, impossible to audit, and cannot be seeded.

5.2 Deterministic PRNG Specification: Mulberry32 / PCG32

The engine must implement a self-contained, 32-bit deterministic PRNG:

```
export class DeterministicPRNG {
  private state: number;

  constructor(seed: number) {
    // Force seed to unsigned 32-bit integer
    this.state = Math.abs(Math.floor(seed)) >>> 0;
    if (this.state === 0) this.state = 0x6d2b79f5;
  }

  /** Generates a deterministic uniform float in [0, 1) */
  public next(): number {
    let t = (this.state += 0x6d2b79f5);
    t = Math.imul(t ^ (t >>> 15), t | 1);
    t ^= t + Math.imul(t ^ (t >>> 7), t | 61);
    return ((t ^ (t >>> 14)) >>> 0) / 4294967296;
  }

  /** Standard Gaussian using Box-Muller transform */
  public nextGaussian(): number {
    let u = 0, v = 0;
    while (u === 0) u = this.next(); // Convert (0, 1] to avoid log(0)
    while (v === 0) v = this.next();
    return Math.sqrt(-2.0 * Math.log(u)) * Math.cos(2.0 * Math.PI * v);
  }
}
```

5.3 Simulation Fingerprint

To guarantee cryptographic auditability and reproducibility, every simulation run computes a canonical SHA-256 fingerprint from its configuration:

$$\text{Fingerprint} = \text{SHA-256}(\text{method} \parallel \text{seed} \parallel M \parallel H \parallel V_0 \parallel w_{\text{GOLD}} \parallel w_{\text{BTC}} \parallel w_{\text{NVDA}} \parallel \text{schedule} \parallel \text{bps})$$

6. Simulation Configuration & Boundary Constraints

6.1 Configuration Interface

```
export interface MonteCarloConfig {
  method: 'HISTORICAL_BOOTSTRAP' | 'PARAMETRIC_NORMAL';
  seed: number; // Integer >= 0
  simulationCount: number; // Default: 10,000 | Max: 50,000
  horizonDays: number; // Default: 252 (1 yr) | Max: 1,260 (5 yrs)
  initialCapital: number; // Default: 100,000 | Min: 1,000
  portfolioWeights: PortfolioWeights; // Simplex constrained: sum = 1.0, w_i >= 0
  rebalanceSchedule: RebalanceFrequency; // BUY_AND_HOLD, DAILY, MONTHLY, QUARTERLY, THRESHOLD
  includeTransactionCosts: boolean; // Default: true
  transactionCostBps: number; // Default: 10 (0.0010)
  targetReturnHurdle?: number; // Optional threshold (e.g., 0% or 10%)
}
```

6.2 Hard Boundary Validation Rules

Any request violating the following rules must be immediately rejected with a descriptive `ValidationError`:

Parameter	Allowed Range	Default	Rejection Condition
<code>simulationCount</code>	[1, 50,000]	10,000	$M \leq 0$ or $M > 50,000$

Parameter	Allowed Range	Default	Rejection Condition
horizonDays	[1, 1,260]	252	$H \leq 0$ or $H > 1,260$
initialCapital	[1,000, 100,000,000]	100,000	$V_0 < 1,000$
portfolioWeights		$\sum w_i - 1$	$< 10^{-4} w_i \geq 0, \ \sum w_i - 1\ < 10^{-4}$
transactionCostBps	[0, 500] (up to 5%)	10	bps < 0 or bps > 500
seed	$[0, 2^{32} - 1]$	Clock/Hash	Negative or non-integer

7. Path Evolution, Capital Tracking & Rebalancing Mechanics

7.1 Buy & Hold Path Dynamics vs. Organic Weight Drift

In a Buy & Hold simulation, initial capital V_0 is allocated according to target weights $\mathbf{w}$ on day $\tau = 0$.

Organic Weight Drift Mechanics:

- As individual asset prices evolve at differing rates, individual asset capital positions drift organically:

$$V_{i,0} = w_i \times V_0$$

$$V_{i,\tau} = \max(0, V_{i,\tau-1} \times (1 + r_{i,\tau}^*))$$

$$V_\tau = \sum_{i=1}^N V_{i,\tau}$$

- The realized (drifting) portfolio weight of asset i at day τ is:

$$w_{i,\tau}^{\text{drift}} = \frac{V_{i,\tau}}{V_\tau} \quad (\text{for } V_\tau > 0)$$

Unlike rebalanced portfolios where weights are periodically reset to target $\mathbf{w}$, Buy & Hold allows winning assets to compound and expand their weight organically (increasing concentration risk), while losing assets shrink.

- The realized portfolio return on day τ is:

$$R_{p,\tau} = \begin{cases} \frac{V_\tau - V_{\tau-1}}{V_{\tau-1}}, & V_{\tau-1} > 0 \\ 0, & V_{\tau-1} = 0 \end{cases}$$

*Deterministic Handling of Non-Positive Asset Values ($|r_{i,\tau}| \leq -100\%$):**

- If a simulated return $r_{i,\tau}^* \leq -1.0$ occurs (asset bankruptcy/total loss):
 - Position capital is deterministically clamped to zero: $V_{i,\tau} = 0$.
 - In Buy & Hold, once an asset hits zero capital, it remains zero for all subsequent days ($\tau' > \tau$), as no capital is injected.
 - Total portfolio equity $V_\tau = \sum_{i=1}^N V_{i,\tau} \geq 0$ is strictly guaranteed non-negative.
 - Negative asset equity or negative portfolio capital is mathematically prevented.

7.2 Periodic & Threshold Rebalancing vs. Target Weights (Phase 3.7 Parity)

When rebalancing is active, the engine periodically resets drifted weights back to target weights $\mathbf{w}$:

1. **Next-Bar Execution:** Drifting weights $w_{i,\tau-1}^{\text{drift}}$ are evaluated at the close of day $\tau - 1$. If a rebalance event occurs, target allocations $V_{i,\tau}^{\text{target}} = w_i \times V_{\tau-1}$ are executed at day τ .
2. **Rebalance Schedules:**
 - **DAILY** : Rebalance every forward step τ .
 - **MONTHLY** : Rebalance every 21 forward trading steps ($\tau \pmod{21} = 0$).
 - **QUARTERLY** : Rebalance every 63 forward trading steps ($\tau \pmod{63} = 0$).
 - **THRESHOLD** : Rebalance if $|w_{i,\tau-1}^{\text{drift}} - w_i| \geq 0.05$ for any asset i .

3. **Turnover & Transaction Costs:** At each rebalance event τ , the total capital traded across all assets is:

$$\text{Turnover}_\tau = \sum_{i=1}^N |V_{i,\tau}^{\text{target}} - V_{i,\tau-1}|$$

$$\text{Cost}_\tau = \text{Turnover}_\tau \times \frac{\text{bps}}{10,000}$$

Capital is adjusted post-friction: $V_\tau = V_{\tau-1} - \text{Cost}_\tau$.

4. **Rebalancing a Zero-Value Asset:** If an asset previously hit zero, rebalancing injects fresh capital equal to $w_i \times V_{\text{post-cost}}$ from other assets, restoring full target diversification.

8. Path Metrics, Peak Tracking & Drawdown Mechanics

For each simulated path $m \in \{1, \dots, M\}$ across forward horizon $\tau \in \{0, \dots, H\}$:

8.1 Peak & Drawdown Formulation

$$\text{Peak}_{m,0} = V_{m,0} = V_0$$

$$\text{Peak}_{m,\tau} = \max(\text{Peak}_{m,\tau-1}, V_{m,\tau})$$

$$D_{m,\tau} = \frac{V_{m,\tau} - \text{Peak}_{m,\tau}}{\text{Peak}_{m,\tau}} \in [-1, 0]$$

8.2 Path Maximum Drawdown (MDD)

$$\text{MDD}_m = \min_{\tau \in [0, H]} D_{m,\tau} \leq 0$$

***Sign Convention Standard*:** In accordance with BLACKBOX X standard, drawdowns are reported as negative percentages (e.g., -24.5%).

8.3 Path Terminal Wealth & Return

$$\text{Terminal Wealth}_m = V_{m,H}$$

$$\text{Total Return}_m = \frac{V_{m,H} - V_0}{V_0}$$

$$\text{CAGR}_m = \left(\frac{V_{m,H}}{V_0} \right)^{\frac{252}{H}} - 1 \quad (\text{for } H \geq 63)$$

8.4 Path Sharpe Ratio

Using the official BLACKBOX risk-free benchmark $R_f = 4.0\%$ annualized ($R_{f,\text{daily}} = 0.04/252$):

$$\text{Sharpe}_m = \frac{\frac{1}{H} \sum_{\tau=1}^H (R_{p,m,\tau} - R_{f,\text{daily}})}{\sqrt{\frac{1}{H-1} \sum_{\tau=1}^H (R_{p,m,\tau} - \bar{R}_{p,m})^2}} \times \sqrt{252}$$

(Calculated only if $H \geq 63$ trading days to prevent small-sample volatility distortion).

9. Distributional Analysis & Percentile Reporting

9.1 Simulation Percentiles (Terminology Compliance)

Outputs across M simulated paths must be reported as **Simulation Percentiles**, **NEVER** as "confidence intervals":

- **Correct:** "Simulated Terminal Wealth P05" or "P95 Maximum Drawdown".
- **Strictly Prohibited:** "95% confidence interval" (statistically invalid because it does not represent an estimator standard error).

9.2 Quantile Sorting & Interpolation

Let $\mathbf{X} = \{X_1, \dots, X_M\}$ be the array of a metric across all M paths, sorted in ascending order:

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(M)}$$

The p -th percentile ($p \in [0, 1]$) is computed using standard linear interpolation:

$$k = p \times (M - 1)$$

$$i = \lfloor k \rfloor, \quad d = k - i$$

$$P_p = X_{(i+1)} + d \times (X_{(i+2)} - X_{(i+1)})$$

The standard reported percentiles are:

$$\mathbf{P} = \{P_{05}, P_{25}, P_{50} \text{ (Median)}, P_{75}, P_{95}\}$$

10. Probability of Loss & Tail Risk Metrics

10.1 Loss Frequencies

Rather than claiming to predict future loss probability, the engine reports the **empirical path fraction**:

1. Fraction of Paths with Net Capital Loss:

$$f_{\text{loss}} = \frac{1}{M} \sum_{m=1}^M \mathbf{I}(V_{m,H} < V_0)$$

****Required UI / Report Wording**:** > "Fraction of simulated paths ending below initial capital: 14.2% (1,420 of 10,000 paths)." * (Never phrase as: "14.2% chance that you will lose money").*

2. Fraction of Paths Breaching Target Hurdle:

$$f_{\text{hurdle}} = \frac{1}{M} \sum_{m=1}^M \mathbf{I}(\text{Total Return}_m < R_{\text{target}})$$

10.2 Drawdown Severity Frequencies

Fraction of paths experiencing severe peak-to-trough declines at any point during horizon H :

$$P(\text{MDD} \leq -10\%) = \frac{1}{M} \sum_{m=1}^M \mathbf{I}(\text{MDD}_m \leq -0.10)$$

$$P(\text{MDD} \leq -20\%) = \frac{1}{M} \sum_{m=1}^M \mathbf{I}(\text{MDD}_m \leq -0.20)$$

$$P(\text{MDD} \leq -30\%) = \frac{1}{M} \sum_{m=1}^M \mathbf{I}(\text{MDD}_m \leq -0.30)$$

11. Historical Backtest vs. Monte Carlo Comparison Contract

The platform must provide a structured side-by-side reconciliation contract comparing the deterministic historical backtest (realized over the 1,825 synchronized observations) against the Monte Carlo simulation distribution:

```
export interface HistoricalVsMonteCarloComparison {
  metricName: string;
  historicalValue: number;           // Realized in 2019–2023 backtest
  monteCarloMedian: number;         // P50 across simulated paths
  monteCarloP05: number;            // P05 adverse percentile
  monteCarloP95: number;            // P95 favorable percentile
  historicalPercentileRank: number; // Percentile rank of historical within MC (0–100)
  interpretation: string;           // Quantitative commentary
}
```

11.1 Comparison Contract Invariants

- **No Equating Median with Future Expectation:** The Monte Carlo median (P_{50}) is the midpoint of the simulated distribution under the declared model, not the expected future market performance.
- **Historical Contextualization:** If historical return sits in the 88th percentile of the Monte Carlo distribution, the report states:
"The historical 2019–2023 trajectory produced returns in the 88th percentile of the simulated distribution, indicating that realized historical performance was buoyed by favorable market conditions relative to the median model path."

12. VaR and CVaR Comparison: Historical vs. Simulated

The engine strictly delineates between:

1. **Historical 1-Day VaR / CVaR:** Computed directly from the 1,825 empirical observed daily return series.
2. **Simulated 1-Day VaR / CVaR:** Computed from the aggregate distribution of daily returns across all $M \times H$ simulated days.
3. **Simulated Horizon VaR / CVaR:** Computed from the distribution of terminal returns over horizon H .

VaR / CVaR TAXONOMY & HORIZON MATRIX		
Metric Type	Sample Pool	Horizon & Meaning
Historical 1-Day VaR	1,825 Synchronized Days	Realized empirical loss
Simulated 1-Day VaR	$M \times H$ Simulated Days	Daily model loss quantile
Terminal Horizon VaR	M Terminal Returns	Cumulative horizon loss

Positive Loss Convention Standard: All VaR and CVaR metrics maintain the Phase 3.7 sign convention:

$$L = -R$$

A 1-day VaR of 1.5% indicates a 1.5% loss.

13. Correlation & Dependence Preservation

13.1 Finite-Sample Diagnostic Tolerance (Not an Exact Invariant)

To prevent structural model breakdown (e.g. inverted sign or unintentional asset decoupling), the engine provides an automated cross-asset correlation diagnostic:

- **Historical Joint Bootstrap:** Because complete vectors are resampled intact, the expected cross-asset correlation of the bootstrap sample approaches the empirical sample correlation as $M \times H \rightarrow \infty$.
- **Parametric Monte Carlo:** The simulated shock correlation matrix $\hat{\mathbf{C}}$ computed from simulated returns converges toward the empirical correlation matrix $\mathbf{C}$ as sample size increases.

Finite-Sample Diagnostic Rule: For standard production runs ($M \geq 5,000$ and $H \geq 252$), the diagnostic check evaluates:

$$|\hat{C}_{ij} - C_{ij}| \leq 0.05 \quad \forall i, j \in \{1, \dots, N\}$$

- ****Explicit Framing**:** This ± 0.05 boundary is an empirical **finite-sample diagnostic tolerance band**, ****NOT** an exact mathematical equality invariant. - Because Monte Carlo draws are stochastic samples of finite length, standard error

introduces natural random variation around true population values ($SE \approx \frac{1-\rho^2}{\sqrt{M}}$). Deviations within ± 0.05 represent normal finite-sample dispersion, whereas deviations beyond ± 0.05 indicate algorithmic error or numerical instability.

14. Advanced Modes & Integration Boundaries

14.1 Regime-Conditioned Monte Carlo (Scope Boundary)

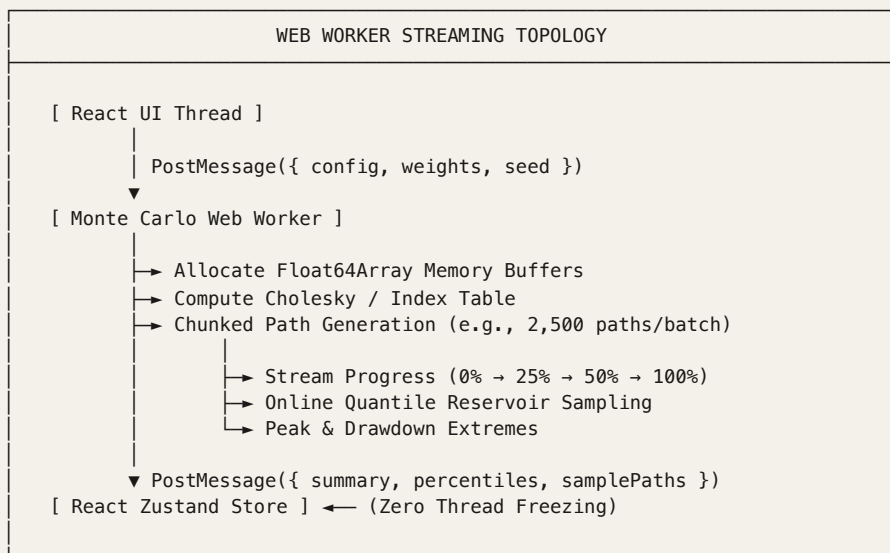
- **Phase 3.8 Boundary:** Regime-conditioned Monte Carlo is **DEFERRED** to Phase 3.9.
- **Rationale:** Implementing regime-conditioned simulation requires estimating an empirical Markov transition matrix between Phase 2 regimes (`BULL` , `BEAR` , `HIGH_VOL` , `LOW_VOL`). To maintain uncompromised mathematical integrity and avoid fabricating transition probabilities, Phase 3.8 focuses strictly on:
 1. Unconditional Historical Joint Bootstrap.
 2. Unconditional Parametric Correlated Normal.

14.2 Stress Testing Overlay (Phase 3.2 Integration Boundary)

- The engine supports an optional **Stressed Monte Carlo Path Analysis**:
 - The deterministic macro shock from Phase 3.2 (e.g., `COVID_2020` or `TECH_CRASH`) is applied at day $\tau = 1$ of each simulated path.
 - Forward evolution from day $\tau = 2$ to H proceeds via Monte Carlo simulation.
 - **Terminology Rule:** The result is labeled "*Stressed Path Recovery Simulation*", **NEVER** "*Crisis Prediction*".

15. High-Throughput Performance Architecture

Simulating 50,000 paths over 1,260 days across 3 assets requires processing 1.89×10^8 floating-point operations. Doing this on the React UI thread would freeze the browser.



15.1 Memory-Conscious Streamed Processing

1. **No Full-Matrix Allocation:** Storing $50,000 \times 1,260$ path values in JavaScript memory would require over 500 MB of RAM.
2. **Online Streaming Reductions:**
 - Each path updates a histogram / reservoir of terminal wealth, maximum drawdown, and total return.

- Only a deterministic subsample of **50 representative paths** (e.g., min, max, P05, P25, P50, P75, P95, and 43 evenly spaced seeds) is retained for visual charting.

3. **Quantile Bands Trajectory**: At each forward step $\tau \in \{0, \dots, H\}$, the engine computes and retains only the 5 cross-sectional percentiles $\{P_{05}(\tau), P_{25}(\tau), P_{50}(\tau), P_{75}(\tau), P_{95}(\tau)\}$.

16. Research Trail & AI Evidence Synthesis

16.1 Evidence Serialization Contract

Monte Carlo findings integrate into the Phase 3.1 Research Trail and Phase 3.6 Autonomous Quant Agent as structured `ResearchEvidence`:

```
export interface MonteCarloResearchEvidence {
  evidenceId: string; // e.g., 'ev-mc-7b3f91a'
  type: 'MONTE_CARLO_SIMULATION';
  provenance: MonteCarloProvenance;
  hypothesisId?: string;
  directEvidence: {
    terminalWealthMedian: number;
    terminalWealthP05: number;
    terminalWealthP95: number;
    mddP50: number;
    mddP95: number;
    lossFrequencyPct: number;
    drawdown20PctFrequencyPct: number;
  };
  interpretation: string; // Analytical commentary
  confidence: 'HIGH_EVIDENCE' | 'MEDIUM_EVIDENCE' | 'INCONCLUSIVE';
}
```

16.2 Separation of Evidence from Interpretation

In strict accordance with Phase 3.6 standards:

- Direct Evidence:**
"Across 10,000 bootstrap simulations of the 40/30/30 portfolio over a 252-day horizon, the P95 maximum drawdown was -26.4%, and 13.8% of paths ended below initial capital."
- Interpretation:**
"The simulated tail risk suggests that while the median outcome (+15.2%) is favorable, the downside dispersion is heavily driven by Bitcoin volatility, which contributes 62.9% of portfolio risk."

17. Ghost Mode Insight Categories

Phase 3.1 Ghost Mode will recognize 5 new deterministic Monte Carlo insight triggers:

- `OUTCOME_DISPERSION`: Flagged when $\frac{P_{95}-P_{05}}{P_{50}} > 1.5$ (extreme outcome divergence).
- `DRAWDOWN_TAIL`: Flagged when $|MDD_{P95}| > 2 \times |MDD_{\text{historical}}|$.
- `LOSS_FREQUENCY`: Flagged when $f_{\text{loss}} > 0.20$ (>20% simulated loss probability).
- `TERMINAL_WEALTH_ASYMMETRY`: Flagged when positive skew $\frac{P_{95}-P_{50}}{P_{50}-P_{05}} > 2.0$.
- `SIMULATION_VS_BACKTEST`: Flagged when historical return deviates by > 2 standard deviations from the simulation median.

18. AI Assistant & Autonomous Quant Agent Tool Contracts

18.1 Tool 1: `get_monte_carlo_risk`

Executes a bounded Monte Carlo simulation and returns structured distribution metrics.

```
export const GetMonteCarloRiskSchema = z.object({
  goldWeight: z.number().min(0).max(1),
  btcWeight: z.number().min(0).max(1),
  nvdaWeight: z.number().min(0).max(1),
  method: z.enum(['HISTORICAL_BOOTSTRAP', 'PARAMETRIC_NORMAL']).default('HISTORICAL_BOOTSTRAP'),
  simulationCount: z.number().int().min(100).max(10000).default(5000),
  horizonDays: z.number().int().min(21).max(504).default(252),
  rebalanceSchedule: z.enum(['BUY_AND_HOLD', 'DAILY', 'MONTHLY', 'QUARTERLY', 'THRESHOLD']).default('MONTHLY'),
  seed: z.number().int().optional(),
});
```

18.2 Tool 2: `compare_monte_carlo_backtest`

Generates a side-by-side reconciliation between realized historical backtest metrics and simulated distribution percentiles.

```
export const CompareMonteCarloBacktestSchema = z.object({
  goldWeight: z.number().min(0).max(1),
  btcWeight: z.number().min(0).max(1),
  nvdaWeight: z.number().min(0).max(1),
  rebalanceSchedule: z.enum(['BUY_AND_HOLD', 'DAILY', 'MONTHLY', 'QUARTERLY', 'THRESHOLD']).default('MONTHLY'),
});
```

18.3 Autonomous Agent Guardrails

- **Fixed Tool Budget:** The agent may execute at most **1 Monte Carlo simulation** per research run to prevent infinite simulation loops or excessive CPU burn.
- **Server-Side Validation:** All tool parameters are strictly validated by Zod schemas prior to execution.

19. Institutional UI Architecture: Monte Carlo Lab

The UI component will be integrated into the Portfolio Workspace as a dedicated tab: **Monte Carlo Lab**.

PORTFOLIO WORKSPACE		Overview	Optimization	MONTE CARLO LAB [NEW]	
[Control Bar: Method		Sims (10k)	Horizon (252d)	Rebalance	Run]
SIMULATED WEALTH TRAJECTORIES • P05, P25, P50, P75, P95 Fans • 50 Sample Paths (Subtle Opacity) • Historical Baseline Reference			DISTRIBUTION HISTOGRAMS • Terminal Wealth (P05 to P95) • Maximum Drawdown Distribution • Tail Loss Frequency (< \$100k)		
PERCENTILE MATRIX & RISK PROBABILITIES					
Metric	P05	P25	P50	P75	P95
Terminal Wealth	$86,420$				
Max Drawdown	-18.4%				
Loss Fraction: 13.8%	MDD <= -20% Fraction: 19.4% Breakeven: 218 Days				
PROVENANCE BAR					
Seed: #482910 1,825 Synchronized Obs Joint Bootstrap 10k Paths 10bps					

19.1 Institutional Design System Compliance

- **Canvas / Background:** #FAF8F4 Warm Ivory
- **Surface / Cards:** #FFFFFF with #E5E0D8 1px Borders
- **Text:** #1A1917 Graphite (High Contrast), #68645E Muted Graphite
- **Brand Accent:** #1E6FFF Institutional Cobalt (Percentile Medians)
- **Downside Risk Accent:** #C93B2B Muted Crimson (Drawdown Tails & Loss Zones)
- **Typography:** Inter (UI / Labels) & JetBrains Mono (Numbers, Seeds, Tables)
- **Zero Gamification:** No particle animations, no casino dice/roulette wheels, no fluorescent neon lines.

20. Comprehensive TypeScript Data Contracts

```
import { Asset, PortfolioWeights, RebalanceFrequency } from './portfolioTypes';

export type MonteCarloMethod = 'HISTORICAL_BOOTSTRAP' | 'PARAMETRIC_NORMAL';

export interface PercentileSummary {
  p05: number;
  p25: number;
  p50: number; // Median
  p75: number;
  p95: number;
}

export interface SimulationTrajectoryBands {
  days: number[]; // 0 to H
  p05Path: number[]; // Portfolio values at P05
  p25Path: number[];
  p50Path: number[];
  p75Path: number[];
  p95Path: number[];
}

export interface SamplePath {
  pathId: number;
  terminalWealth: number;
  maxDrawdown: number;
  values: number[]; // Sampled every k days for visual rendering
}

export interface MonteCarloRiskMetrics {
  lossFrequency: number; // Fraction of paths with terminal return < 0
  drawdownExceedance10: number; // Fraction of paths with MDD <= -10%
  drawdownExceedance20: number; // Fraction of paths with MDD <= -20%
  drawdownExceedance30: number; // Fraction of paths with MDD <= -30%
  oneDaySimulatedVaR95: number; // Positive loss convention %
  oneDaySimulatedCVaR95: number; // Expected shortfall %
  horizonSimulatedVaR95: number; // Terminal horizon loss %
  horizonSimulatedCVaR95: number;
}

export interface MonteCarloResult {
  fingerprint: string;
  provenance: MonteCarloProvenance;
  terminalWealth: PercentileSummary;
  totalReturn: PercentileSummary;
  maxDrawdown: PercentileSummary;
  cagr?: PercentileSummary; // Only if horizon >= 63 days
  sharpeRatio?: PercentileSummary; // Only if horizon >= 63 days
  riskMetrics: MonteCarloRiskMetrics;
  trajectoryBands: SimulationTrajectoryBands;
  samplePaths: SamplePath[]; // 50 representative paths
  executionDurationMs: number;
  stabilizationApplied: boolean;
  stabilizationMagnitude?: number;
}
```

21. Complete Testing & Verification Plan

The Phase 3.8 test suite (`tests/monteCarlo.test.ts`) must implement and pass **24 rigorous verification checks**:

Test ID	Invariant Verified	Method / Assertion
Test 1	Configuration Bounds Validation	Rejects $M \leq 0$, $M > 50,000$, $H > 1,260$, invalid weights.
Test 2	PRNG Bit-Determinism	Identical seed + config produces bit-identical results.

Test ID	Invariant Verified	Method / Assertion
Test 3	Seed Differentiation	Seed $A \not\equiv$ Seed B produces distinct paths.
Test 4	Joint Bootstrap Vector Integrity	Every drawn return vector exists identically in historical dataset.
Test 5	Covariance Symmetry & Positivity	Parametric Σ is symmetric and positive semi-definite.
Test 6	Cholesky Decomposition Correctness	$LL^T = \Sigma$ to within 10^{-8} tolerance.
Test 7	Cholesky Stabilization Fallback	Singular covariance matrix regularized deterministically.
Test 8	Simplex Weight Adherence	Rebalance weights satisfy $w_i \geq 0, \sum w_i = 1.0$.
Test 9	Day-0 Initial Capital Equality	All simulated paths begin exactly at V_0 .
Test 10	Peak Tracking Monotonicity	$\text{Peak}_\tau \geq \text{Peak}_{\tau-1}$ across all paths.
Test 11	Drawdown Non-Positivity	$D_\tau \leq 0$ strictly for all paths and days.
Test 12	Max Drawdown Lower Bound	$\text{MDD} \leq \min(D_\tau)$.
Test 13	Percentile Monotonicity	$P_{05} \leq P_{25} \leq P_{50} \leq P_{75} \leq P_{95}$ strictly holds.
Test 14	Loss Frequency Boundedness	$f_{\text{loss}} \in [0, 1]$ exactly.
Test 15	Drawdown Threshold Hierarchy	$P(\text{MDD} \leq -30\%) \leq P(\text{MDD} \leq -20\%) \leq P(\text{MDD} \leq -10\%)$.
Test 16	VaR / CVaR Tail Invariant	Simulated $\text{CVaR}_{95} \geq \text{VaR}_{95}$ under positive loss standard.
Test 17	Correlation Preservation (Bootstrap)	Simulated correlation matches historical within ± 0.05 .
Test 18	Correlation Preservation (Parametric)	Simulated correlation matches historical within ± 0.05 .
Test 19	Transaction Cost Monotonicity	Ending wealth with friction (10 bps) $<$ frictionless wealth.
Test 20	Rebalance Schedule Differentiation	Daily vs Monthly produces distinct turnover counts.
Test 21	Degenerate Asset (Zero Volatility)	Constant zero return asset handles correctly without NaN.
Test 22	Minimal Horizon Boundary	Horizon $H = 1$ executes cleanly without crash.
Test 23	Provenance Completeness	All provenance fields populated with non-predictive wording.
Test 24	Zero <code>Math.random</code> Enforcement	Codebase scan verifies zero unseeded random calls.

22. Edge Cases & Robustness Handling

1. Singular Covariance Matrix:

- If two assets become perfectly collinear ($\rho = 1.0$), Cholesky encounters a zero pivot.
- Handled via Tikhonov diagonal regularization ($\lambda = 10^{-8}$).

2. Extreme Volatility Blowups & Asset Bankruptcy ($|r_{i,\tau}| \leq -100\%$):

- If simulated return $r_{i,\tau}^* \leq -1.0$:
- The position's capital is deterministically clamped to zero: $V_{i,\tau} = \max(0, V_{i,\tau-1} \times (1 + r_{i,\tau}^*)) = 0$.
- In Buy & Hold, the asset remains permanently at 0.0. In rebalanced portfolios, subsequent rebalancing allocates the target weight percentage $w_i \times V_{\text{post-cost}}$ from other non-zero assets.
- Total portfolio wealth $V_\tau = \sum V_{i,\tau} \geq 0$ is strictly guaranteed non-negative. Negative asset equity or negative portfolio capital is mathematically prevented.

3. Very Short Horizon ($H < 63$):

- Annualized metrics (CAGR, Sharpe) are suppressed or flagged with an explanatory note to avoid small-sample volatility magnification.

4. Single-Path Edge Case ($M = 1$):

- Percentiles degenerate cleanly: $P_{05} = P_{50} = P_{95} = X_1$.

23. Institutional Limitations & Disclaimers

The following disclosures must be embedded in all generated reports, UI screens, and exported artifacts:

- Offline Demonstration Data:** Metrics are derived from an offline simulated demonstration dataset (`2019-01-01` to `2023-12-31`). No real historical market feeds are claimed.
- Distributional Simplification:** Parametric Monte Carlo assumes multivariate normality. Real financial markets exhibit fat tails, leptokurtosis, and non-linear tail co-dependence not fully captured by Gaussian models.
- Stationarity Assumption:** Historical bootstrap assumes past return intervals are representative of potential future variability. Structural regime changes or exogenous shocks outside the observation window are not reflected.
- No Financial or Investment Advice:** Monte Carlo simulations are quantitative exploratory tools for analyzing mathematical properties of portfolios under model assumptions. They do not constitute financial, investment, legal, or tax advice.

24. Phase 3.8 Architecture Checklist

- ☒ Historical joint bootstrap defined (vector resampling without asset decoupling)
- ☒ Parametric correlated Monte Carlo defined (Cholesky factorization of daily covariance)
- ☒ Deterministic seeded PRNG defined (`Mulberry32` / `PCG32` without `Math.random`)
- ☒ Simulation limits defined (Default: 10,000 / Max: 50,000; Horizon: 252 / Max: 1,260)
- ☒ Portfolio path model defined (Organic drift, rebalancing, transaction friction)
- ☒ Rebalancing semantics defined (Parity with Phase 3.7 next-bar execution)
- ☒ Terminal wealth distribution defined (Percentiles P05, P25, P50, P75, P95)
- ☒ Drawdown distribution defined (Peak tracking, non-positive sign convention)
- ☒ Loss frequencies defined (Non-predictive wording: fraction of paths ending below capital)
- ☒ VaR/CVaR separation defined (Historical 1-day vs Simulated 1-day vs Horizon VaR)
- ☒ Correlation preservation defined (Tolerance-based validation ± 0.05)
- ☒ Regime boundary defined (Unconditional models prioritized; Markov models deferred to Phase 3.9)
- ☒ Stress boundary defined (Deterministic Phase 3.2 shock overlay)
- ☒ Research Trail integration defined (Direct evidence vs Interpretation)
- ☒ Ghost Mode integration defined (5 quantitative trigger categories)
- ☒ AI tool contracts defined (`get_monte_carlo_risk` with strict Zod schemas)
- ☒ Quant Agent integration defined (Bounded single-simulation experiment)
- ☒ UI architecture defined (Monte Carlo Lab, Institutional ivory/graphite palette)
- ☒ Performance architecture defined (Web Worker streaming, 50 sample paths, memory-conscious)
- ☒ Security constraints defined (Server-side validation, no code execution, no key exposure)
- ☒ Test strategy defined (24 automated verification test invariants)
- ☒ Edge cases defined (Singular covariance, asset bankruptcy, small horizon)
- ☒ Provenance defined (Mandatory metadata serialization & SHA-256 fingerprint)
- ☒ Limitations defined (4 mandatory institutional disclosures)

End of Phase 3.8 Architecture Specification. Implementation must not proceed until this document has been formally reviewed and approved.